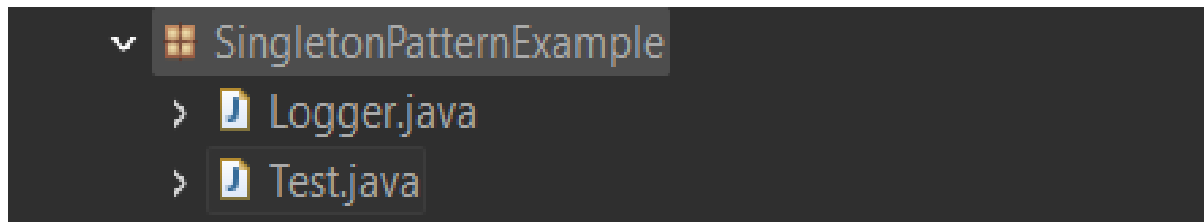


# DESIGN PRINCIPLES AND PATTERNS

## 1.SINGLETON PATTERN EXAMPLE

```
1 package SingletonPatternExample;
2
3 public class Logger {
4
5     private static Logger obj;
6
7     private Logger() {
8
9     }
10
11     public static Logger getInstance() {
12         if(obj == null) {
13             obj = new Logger();
14         }
15
16         return obj;
17     }
18
19 }
20
```

```
1 package SingletonPatternExample;
2
3 public class Test {
4
5     public static void main(String[] args) {
6
7         Logger l1 = Logger.getInstance();
8         Logger l2 = Logger.getInstance();
9         //As we are able to create only one obj --> both the reference variable are same
10        System.out.println(l1 == l2);
11    }
12
13 }
14
```



## 2. FACTORY METHOD PATTERN

```
1 package FactoryMethodPatternExample;
2
3 public interface Document {
4
5     void open();
6
7 }
8
```

```
1 package FactoryMethodPatternExample;
2
3 public class PdfDocument implements Document {
4
5     @Override
6     public void open() {
7         System.out.println("PDF Opened");
8     }
9
10 }
11
```

```
1 package FactoryMethodPatternExample;
2
3 public class WordDocument implements Document{
4
5     @Override
6     public void open() {
7         System.out.println("Word Opened");
8     }
9
10 }
11
```

```
1 package FactoryMethodPatternExample;
2
3 public class ExcelDocument implements Document{
4
5     @Override
6     public void open() {
7         System.out.println("Excel opened");
8     }
9
10
11
12
13 }
```

```

1 package FactoryMethodPatternExample;
2
3 public class DocumentFactory {
4
5     public static Document createDocument(String type) {
6
7         if(type.equals("PDF")) {
8             return new PdfDocument();
9         }
10        else if(type.equals("Word")) {
11            return new WordDocument();
12        }
13
14        return null;
15    }
16
17 }

```

```

1 package FactoryMethodPatternExample;
2
3 public class Test {
4
5     public static void main(String[] args) {
6
7         Document d1 = DocumentFactory.createDocument("PDF");
8         d1.open();
9         Document d2 = DocumentFactory.createDocument("Word");
10        d2.open();
11    }
12 }
13

```

```

v  FactoryMethodPatternExample
  > Document.java
  > DocumentFactory.java
  > ExcelDocument.java
  > PdfDocument.java
  > Test.java
  > WordDocument.java

```